

Smart Agriculture Assistant: Multilingual Farmer Chatbot with Voice, Text, and Image Support

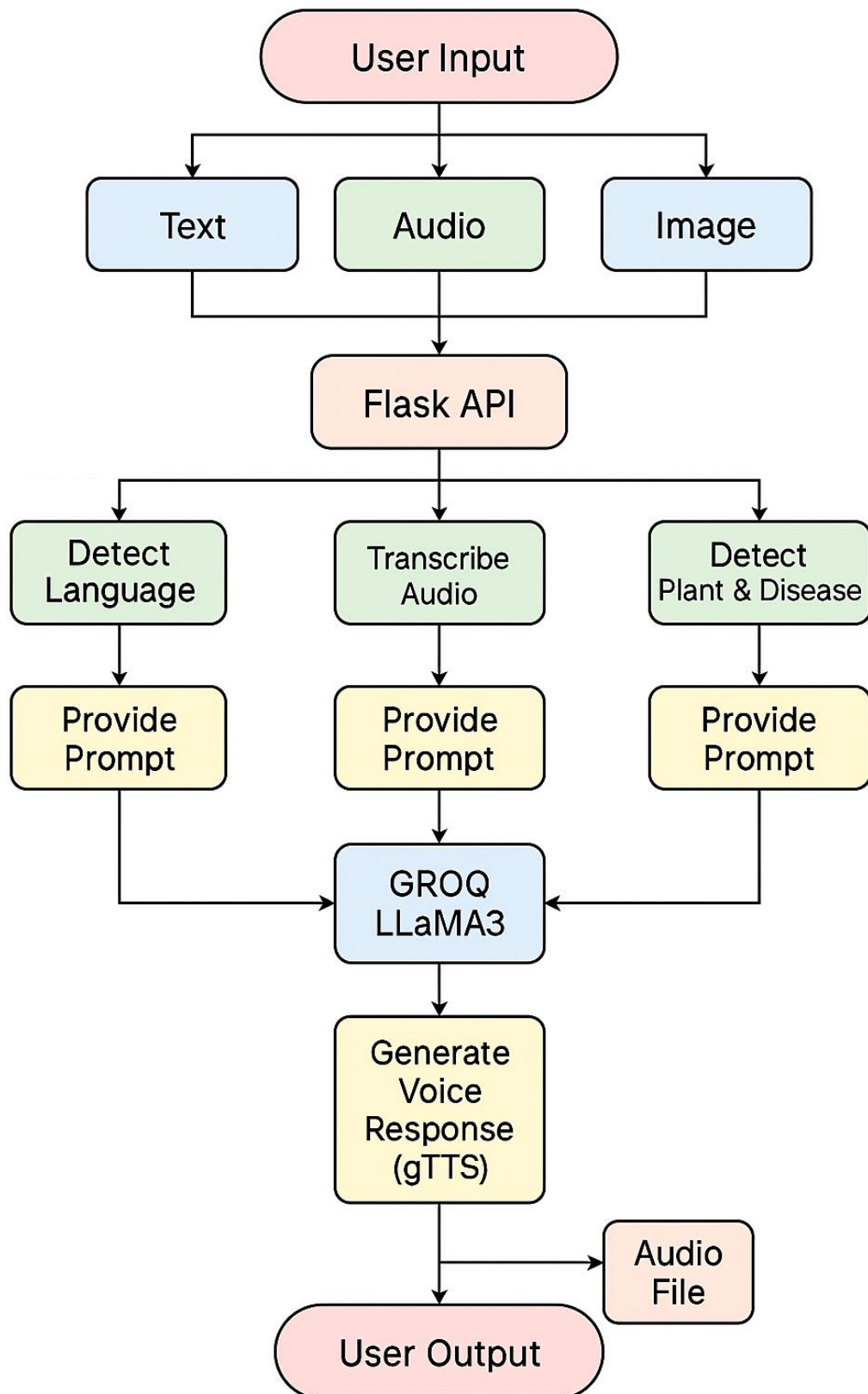
1. Project Overview

- This project aims to build a smart agricultural assistant chatbot that can understand text, voice, and image inputs and respond with useful crop and disease-related advice. It is designed specifically for Indian farmers, supporting Marathi, Hindi, and English languages for both input and output.
- The chatbot leverages YOLO-based deep learning models for plant and disease detection from images, Groq's Whisper for voice transcription, Groq's LLM for language understanding, and Google TTS (gTTS) for converting responses to audio. The system is accessible via a web interface using Flask on the backend and HTML/CSS/JavaScript on the frontend.

2. Key Features

- **Multilingual Support** (English, Hindi, Marathi)
- **Voice Input & Output** using Whisper and gTTS
- **Text-based Q&A Chat Interface**
- **Image Upload for Plant/Disease Detection**
- **Plant and Disease Classification using YOLOv8**
- **Language-specific Output Generation**
- **Audio Playback of Responses**
- **User-friendly Web Dashboard**

Flowchart:



3. Technologies Used

Layer	Technology
Frontend	HTML, CSS, JavaScript (Vanilla)
Backend	Python (Flask Framework)
AI Models	Groq API (LLM, Whisper)
TTS	Google Text-to-Speech (gTTS)
Vision	YOLOv8 via Ultralytics
Language Detection	langdetect

4. Libraries & Dependencies

- **Flask** – for web app routing and server handling
- **ultralytics** – to run YOLOv8 object detection models
- **gTTS** – to generate speech from text
- **groq** – API client for Whisper and LLM (LLaMA3)
- **torch** – PyTorch backend used by YOLO
- **langdetect** – for auto-detecting user input language
- **werkzeug** – secure file uploads
- **re** – for text cleaning and formatting
- **random, string, os** – Python standard utilities

5. Model Details

- **plant-identification.pt:** Trained YOLOv8 model for identifying plant types (e.g., Tomato, Cotton, etc.)
- **plant-disease-detector.pt:** Trained YOLOv8 model for detecting common plant diseases (e.g., Alternaria, Leaf Spot, etc.)

6. How it Works:

1. **Text Input:** User enters a farming-related query. The bot detects language, queries Groq LLM, generates a reply, and speaks it.
2. **Voice Input:** User records a voice query. It's transcribed using Groq Whisper, passed to LLM, and answered in the same language.
3. **Image Input:** User uploads an image of a plant leaf. YOLO identifies the plant and disease, then Groq LLM generates a treatment suggestion in selected language.

7. Overall Summary:

This project successfully integrates advanced AI technologies into a single platform to help Indian farmers with real-time solutions to crop problems. The chatbot is easy to use, supports natural communication through voice, and works even with limited text input. It is a practical, scalable solution for precision agriculture and farmer support.